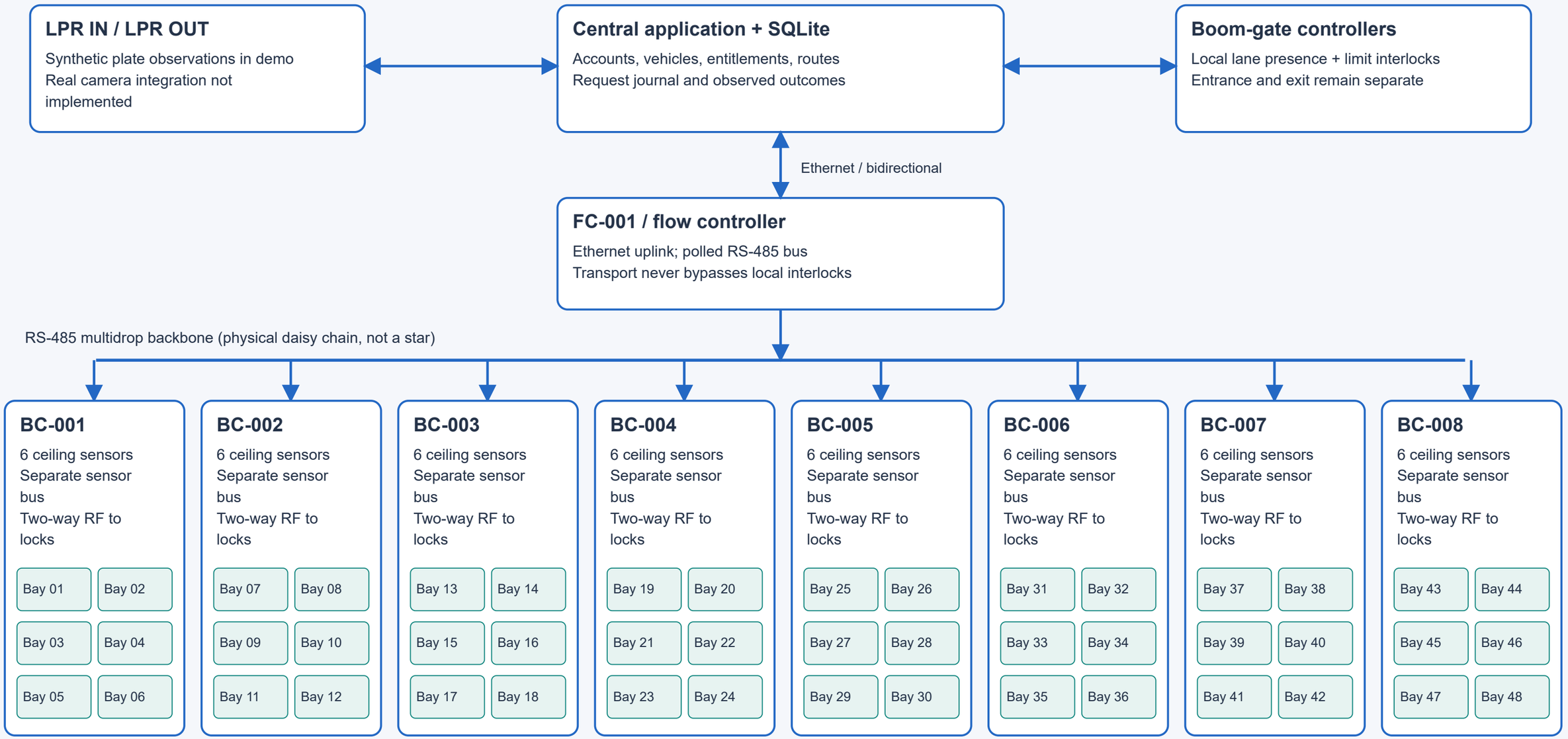
Complete logical hierarchy

1 flow controller / 8 six-bay controllers / 48 planned endpoints. Nine current endpoints are mapped in the central demo.

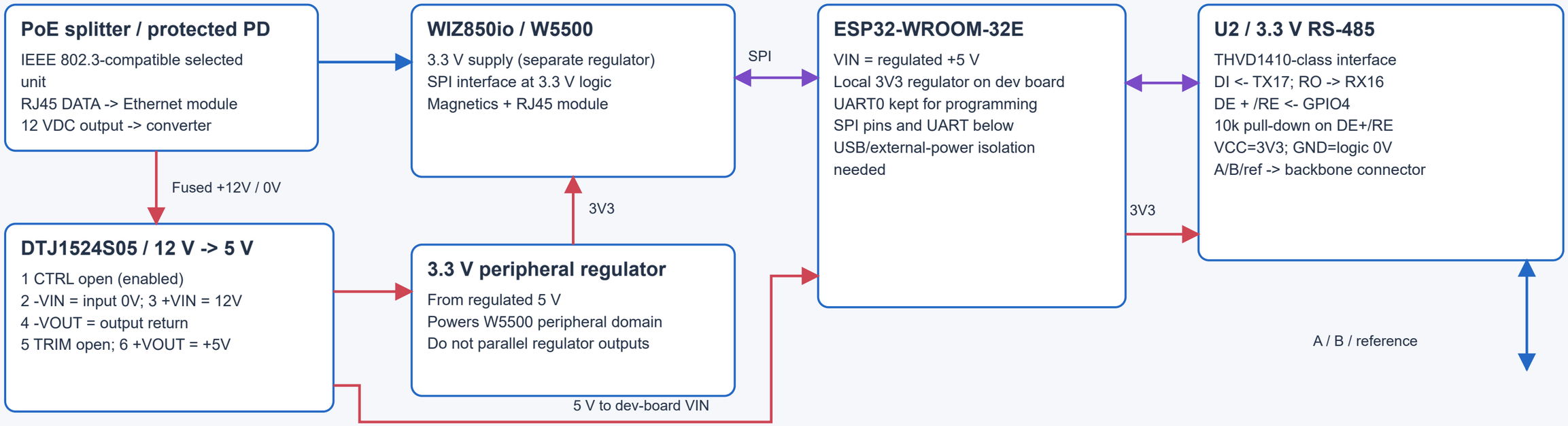


BENCH / MODULE REFERENCE - not a certified safety circuit or fabrication-ready PCB. See wiring schedule and design notes.

GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.

Flow controller - power, Ethernet and RS-485

Pin-complete reference uses an ESP32 DevKitC plus WIZ850io/W5500 module; this is an explicit alternative to the BOM WT32-ETH01.



Converter output return is logic 0V. On an isolated installation, define the reference/isolation scheme explicitly; do not assume PE = signal GND.

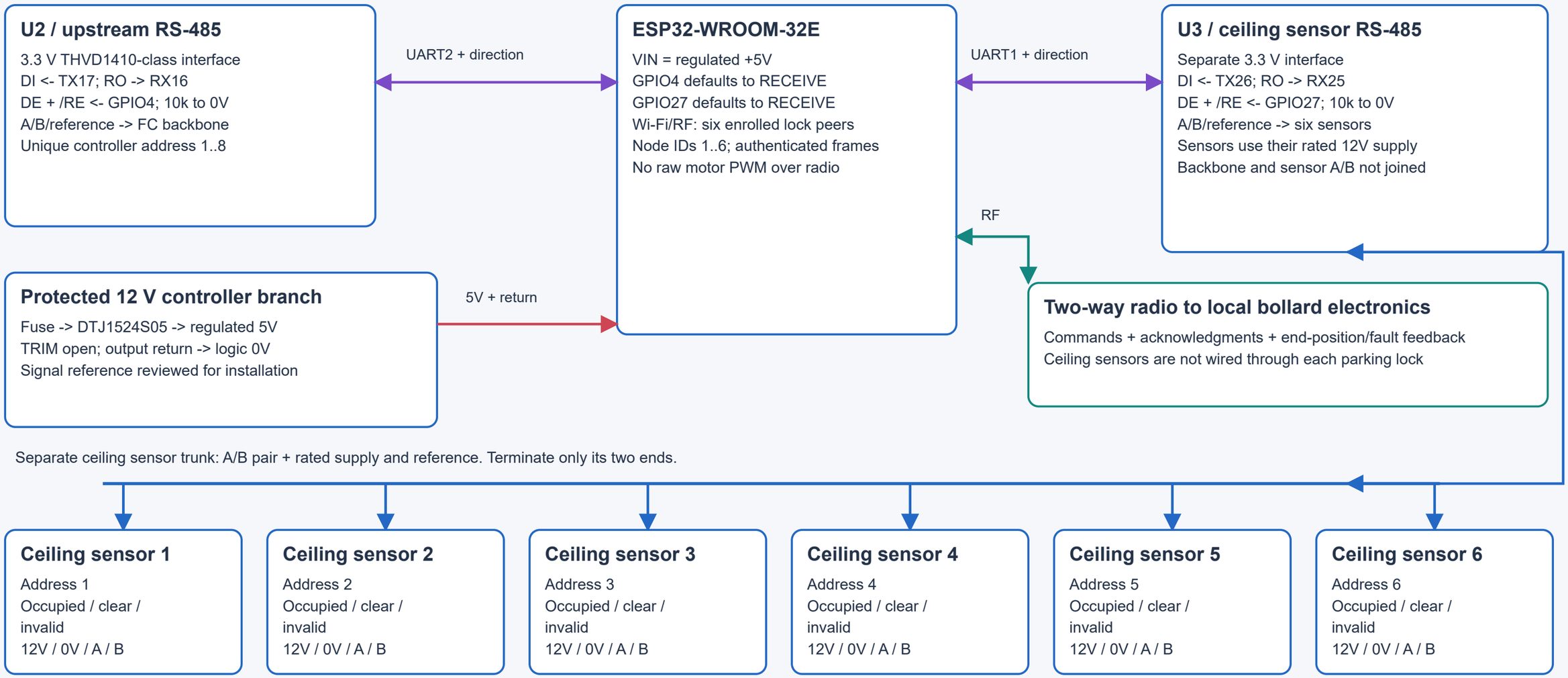
Function	ESP32 GPIO	Module / destination
Ethernet SCLK	18	W5500 SCLK
Ethernet MOSI / MISO	23 / 19	W5500 MOSI / MISO
Ethernet CS / RESET	27 / 21	W5500 /SCS / /RESET (module reset per datasheet)
Ethernet interrupt (optional)	22	W5500 /INT
RS-485 UART TX	17	U2 DI
RS-485 UART RX	16	U2 RO
RS-485 direction	4	U2 DE tied to /RE; 10k to 0V
Bus reference / supply	0V / 3V3	U2 GND / VCC; no 5 V into ESP32 GPIO

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GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.

Six-bay controller - separate upstream and sensor buses

The controller aggregates six ceiling sensors and six addressed bollard nodes. Two independent RS-485 ports require two interfaces.



Short-wire HC-SR04 bench alternative uses the GPIO schedule on Sheet 04, not the industrial sensor bus. Choose one sensor interface mode.

A missing sensor sample or radio heartbeat invalidates a raise permit. Local obstruction and movement limits remain at each bollard.

BENCH / MODULE REFERENCE - not a certified safety circuit or fabrication-ready PCB. See wiring schedule and design notes.

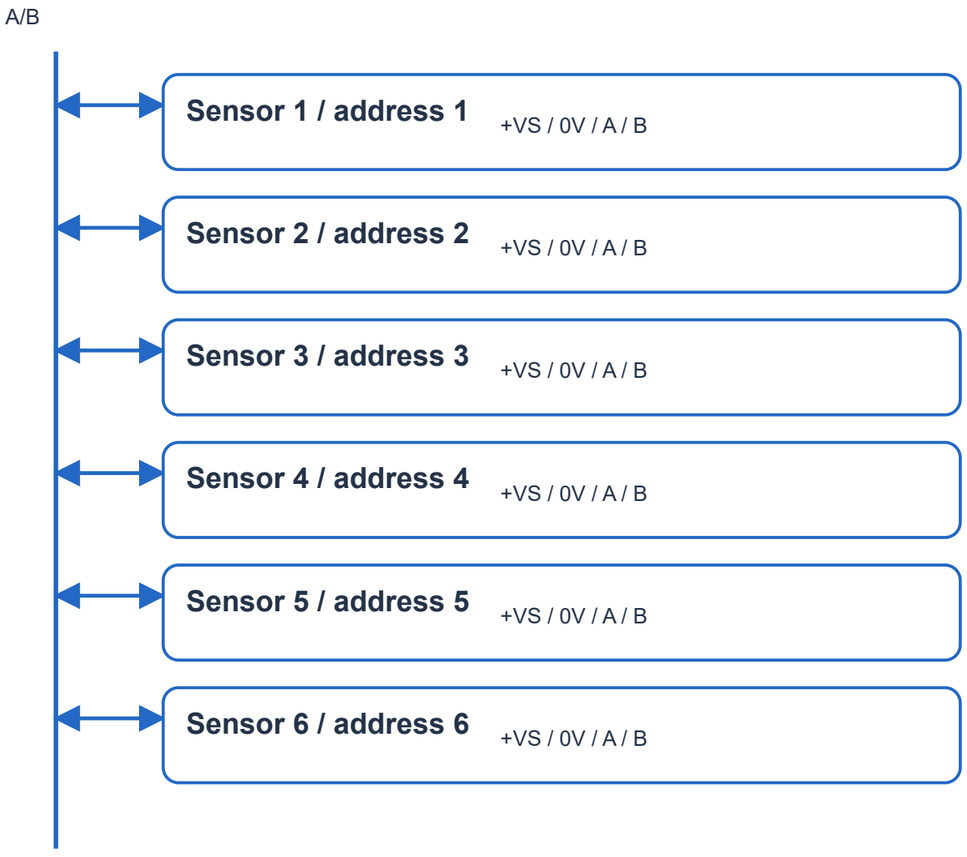
GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.

Ceiling sensing - installation interface and bench alternative

Occupancy is independent from lock position. Commission ceiling-height baselines; do not copy the floor-mounted demo distance thresholds.

A / installation reference: addressed RS-485 sensors

Select sensor model rated for the chosen supply, ceiling height and environment.
Four-wire connectors: +VS, 0V/reference, A, B.
Use a bus trunk with short drops; reserve terminations for its ends.

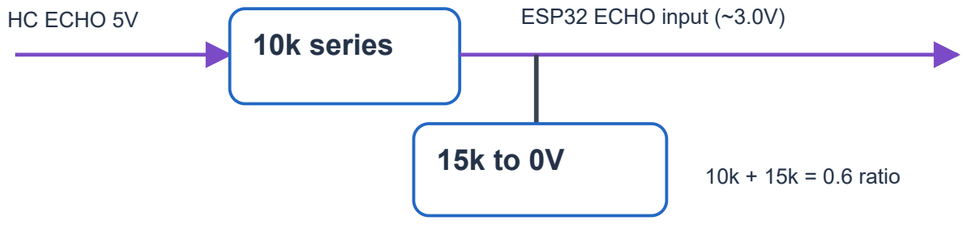


B / bench only: six HC-SR04 units

Short local wires only; not a parking-row cable specification.
VCC=5V; every GND returns to controller logic 0V.
Each trigger uses a 74AHCT125 gate at 5V (2 ICs for 6 gates).
Each 5V echo uses its own 10k/15k divider before ESP32 input.

Sensor	Trigger GPIO (buffered)	Echo GPIO (divided)
1	13	32
2	14	33
3	18	34
4	19	35
5	21	36
6	22	39

Echo network, repeated once per sensor:

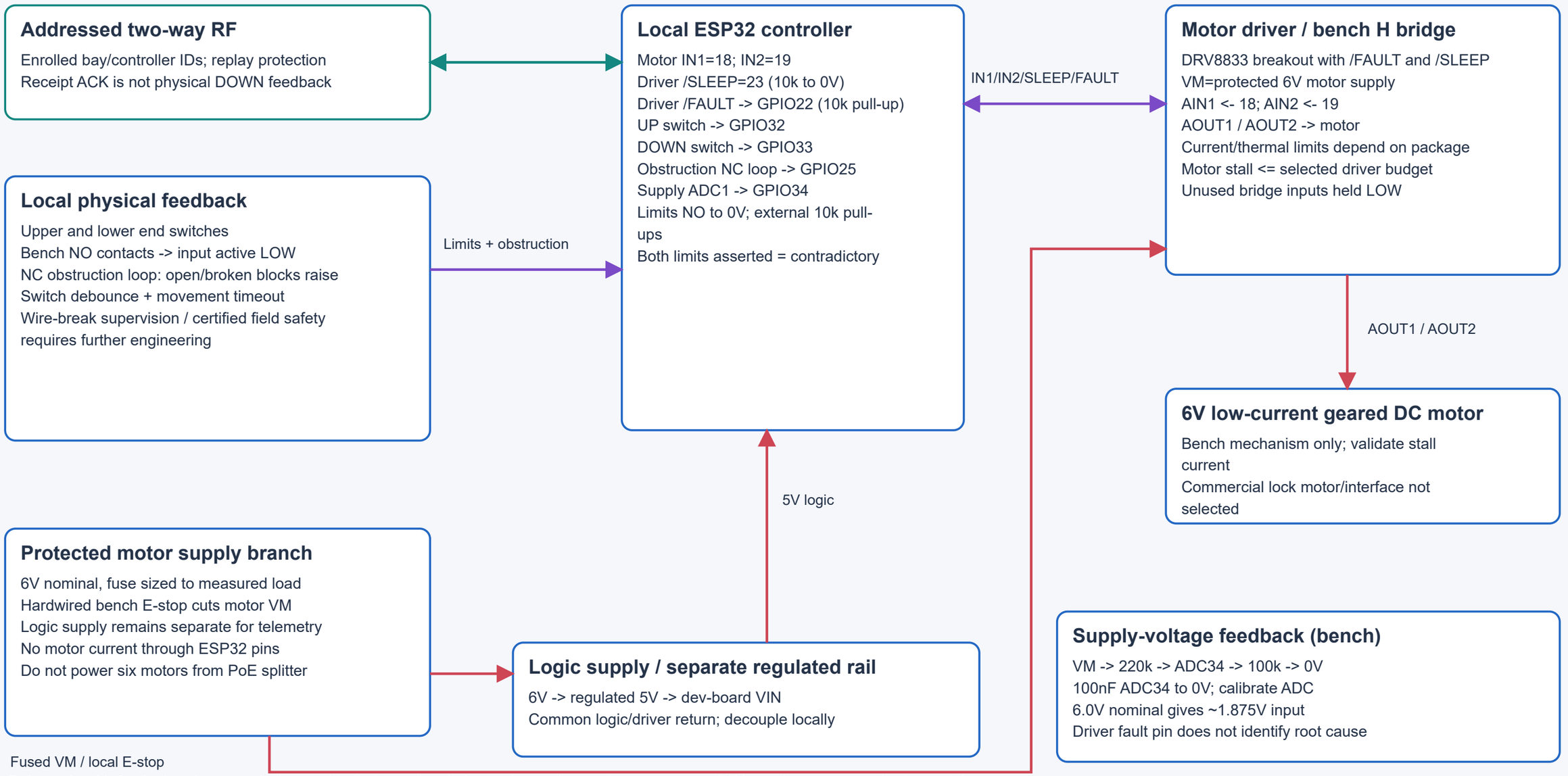


Trigger sensors sequentially to limit acoustic cross-talk; timeout and stale readings mean UNKNOWN.

BENCH / MODULE REFERENCE - not a certified safety circuit or fabrication-ready PCB. See wiring schedule and design notes.
GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.

Local bollard node - actuator, feedback and radio

Custom low-current bench mechanism. The purchased 6V remote lock is NOT assumed to expose this interface or to accept ESP32 radio packets.



Raise only with fresh clear-bay permission, stable limits, no local obstruction and no active emergency. RF loss alone never authorizes raising.

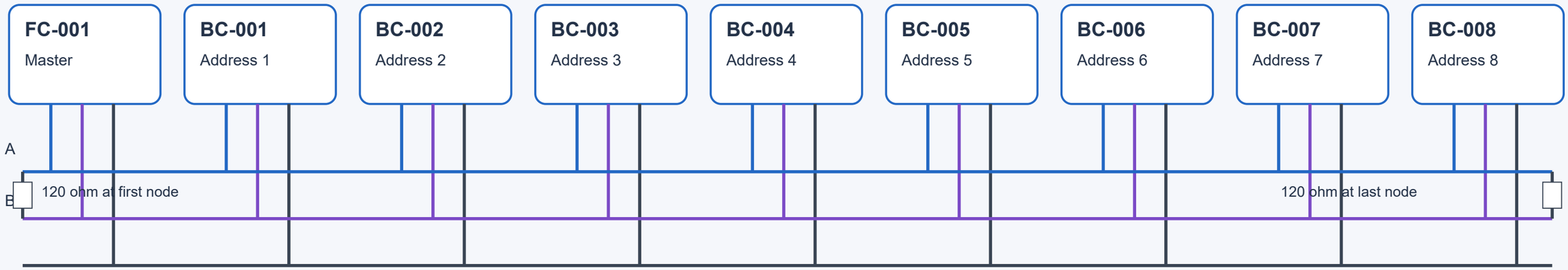
BENCH / MODULE REFERENCE - not a certified safety circuit or fabrication-ready PCB. See wiring schedule and design notes.

GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.

RS-485 wiring - upstream bus and separate sensor bus

A/B must be consistent with the selected module datasheet. RS-485 is an electrical interface; addressing, integrity and freshness belong to the protocol.

MAIN BACKBONE: FC-001 plus BC-001 .. BC-008



Reference conductor: not a motor-current return. Shield / chassis termination requires installation review.

Direction control, grounding and protection

DI <- UART TX; RO -> UART RX
DE and /RE tied together, driven by direction GPIO
10k direction pull-down: receiver enabled at boot
Use 3.3V transceivers with idle/open/short failsafe
THVD1410-class reference needs no external idle bias
MAX485 / MAX3485 substitutions need electrical review
Provide connector ESD protection; assess isolation for field cabling
Do not connect every node with a separate long star cable

Independent ceiling-sensor bus at each BC

BC sensor port -> sensor 1 -> ... -> sensor 6
Do not join its A/B pair to the upstream backbone
Terminate at BC sensor port and final sensor only
Each sensor has a unique address and freshness deadline
Power conductors and fuses follow measured load / voltage drop
Main bus: 9 interfaces. Separate sensor port: 1 interface per BC
Industrial sensors include their own bus interface
Raw HC-SR04 bench mode replaces this sensor bus

Commission one controller with one sensor and one actuator first; then six bays; then eight controller addresses. Scope is design capacity, not measured field reliability.

Central rack, network and power boundaries

Your 8U layout can fit the following allocation only after enclosure depth, equipment height, airflow and mounting are checked. The team BOM lists a 9U cabinet.

U8 / Exhaust / ventilation panel

U7 / CAT6 patch panel

U6 / Managed PoE switch - PoE-capable SKU required

U5 / Router / firewall - restricted control network

U4-U3 / Vented 2U shelf: mini PC / central application / database

U2 / 230V-compatible rack UPS for network / server load

U1 / PDU / socket distribution - matching site voltage

8U allocation: 1 + 1 + 1 + 1 + 2 + 1 + 1 = 8U; no spare space. A 9U cabinet provides one additional rack unit.

Central application: /central | User: /app | Security: /security | Connected circuit: /circuit/ | Native 3D: /?view=three

Data path

LPR cameras + gate controllers -> managed switch
Mini PC serves API, apps and SQLite over Ethernet
PoE/data -> compliant splitter -> flow controller
Flow controller -> RS-485 -> eight bay controllers
Bay controllers -> ceiling sensors and RF lock nodes

Database / demo boundary

public.sqlite: identities, entitlements, controller registry
Core circuit session: runtime memory; reconnect after restart
scaled.sqlite: separate capacity-model demo
Docker named volume retains files across normal restart
SQLite is sufficient for this single-host demo
Hardware access goes through the API, not direct DB writes

Power corrections / pending checks

WS-C2960S-24TS-S in BOM does not provide PoE
SMART500RT1U in BOM is a 120V UPS
Do not connect a 120V unit to a 230V supply
Parking motors require separately sized protected power
Rack UPS runtime does not establish motor backup runtime
Mains protection / bonding are installation design work

Entrance / exit gate controller and LPR interface

One reference controller per lane. Use the gate manufacturer's local safety controller; do not power a barrier motor from ESP32 pins.

LPR camera / simulated plate source

PoE CAT6 -> managed switch -> central API
Central resolves owner / approved visitor
Demo: plate entry form, not camera OCR
Real camera event API and credentials pending

Central network

Lane ESP32 + Ethernet

Same DevKit / W5500 reference as Sheet 02
SPI18/23/19; CS27; RESET21; INT22
OPEN request -> GPIO25
CLOSE request -> GPIO26
Commands require short expiry + replay checks
Watchdog must not replay motion on boot

Logic only

Two normally-open dry contacts

GPIO25/26 -> 3.3V-compatible relay drivers
Active-HIGH drivers; inputs 10k to 0V
Driver supply and flyback per module
NO1 -> vendor OPEN input; COM1 -> input COM
NO2 -> vendor CLOSE input; COM2 -> input COM
Contacts rated for vendor control circuit
No mains or motor power on relay contact plan

Isolated contact requests

Vendor barrier safety controller

OPEN / CLOSE pulses per vendor manual
Controls motor, brakes and travel limits
Safety beam / vehicle loops wired locally
Hardwired STOP per vendor safety design
Controller decides movement / hold-open
No automatic close from stale central data

Vendor status -> conditioned inputs -> GPIO

Isolated / conditioned status inputs

UP limit -> GPIO32; DOWN limit -> GPIO33
Beam healthy / clear -> GPIO34
Vehicle loop clear -> GPIO35
Local stop healthy -> GPIO36
External 10k pull-ups at 3V3 where applicable
24V vendor outputs need isolated interfaces
Input-only pins have no internal pull-ups
Broken or stale status blocks unsafe requests

Observed feedback

Lane-specific interlocks and acceptance

Ceiling bay occupancy is NOT a gate safety beam. Validate lane sensors, passage, limits and stop independently.
Duplicate IDs, power loss, reboot, occupied beam and expired requests must not produce unexpected closure.
Existing Wokwi entry/exit servos visualize motion; they do not validate this real gate relay interface.

Gate power and emergency release follow the vendor installation manual. Confirm dry-contact voltage, pulse timing and status availability before wiring.

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GPIO maps: ESP32-WROOM-32E DevKitC. The team WT32-ETH01 requires its own pin-map review. Motor current and field safety remain unverified.